

Bitcoin Market Sentiment vs Hyperliquid Trader Performance Analysis

Executive Summary

This report analyzes the relationship between Bitcoin market sentiment, measured using the Fear & Greed Index, and trader performance using Hyperliquid trading data. The two datasets were merged based on trading date, resulting in a final dataset of **211,218** trading records after data cleaning.

The analysis examined trader profitability, trading activity, BUY/SELL behavior, and asset performance across different market sentiment categories. The findings indicate that while trading activity was highest during **Fear**, the highest average realized profits were observed during **Extreme Greed**. Additionally, BUY trades performed better during Fear, whereas SELL trades generated higher average profits during Greed and Extreme Greed. These results suggest that market sentiment is associated with changes in trading behavior and realized profitability, although the analysis does not establish a causal relationship.

Methodology

Data Collection

Two datasets were used for this analysis:

- **Hyperliquid Trader Dataset** containing trade-level information such as traded asset, trade direction, execution price, trade size, and realized profit/loss (Closed PnL).
- **Bitcoin Fear & Greed Index Dataset** containing daily Bitcoin market sentiment classified into categories such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed.

Data Preprocessing

The following preprocessing steps were performed:

- Converted timestamp columns into datetime format.
- Extracted the trading date from the timestamp.
- Merged both datasets using the trading date.
- Identified and removed six records that had no matching sentiment data.
- Verified missing values and duplicate records before analysis.

Exploratory Data Analysis

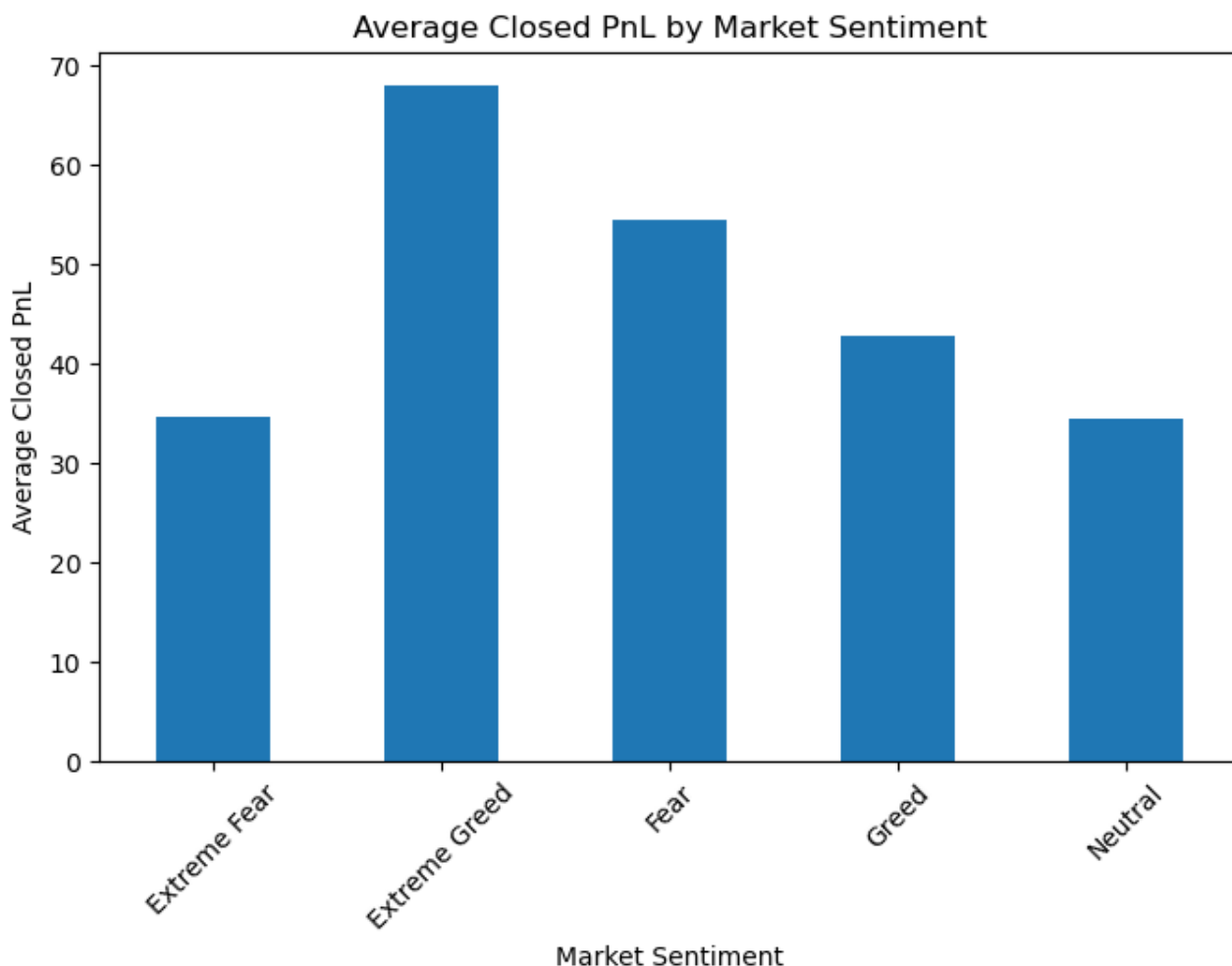
The merged dataset was analyzed to understand:

- Distribution of Closed PnL.
- Trading activity across different market sentiment categories.
- Average trader profitability under different market conditions.
- BUY versus SELL trading behaviour.
- Performance of major traded assets.

Several statistical summaries and visualizations were created to identify meaningful patterns and relationships within the data.

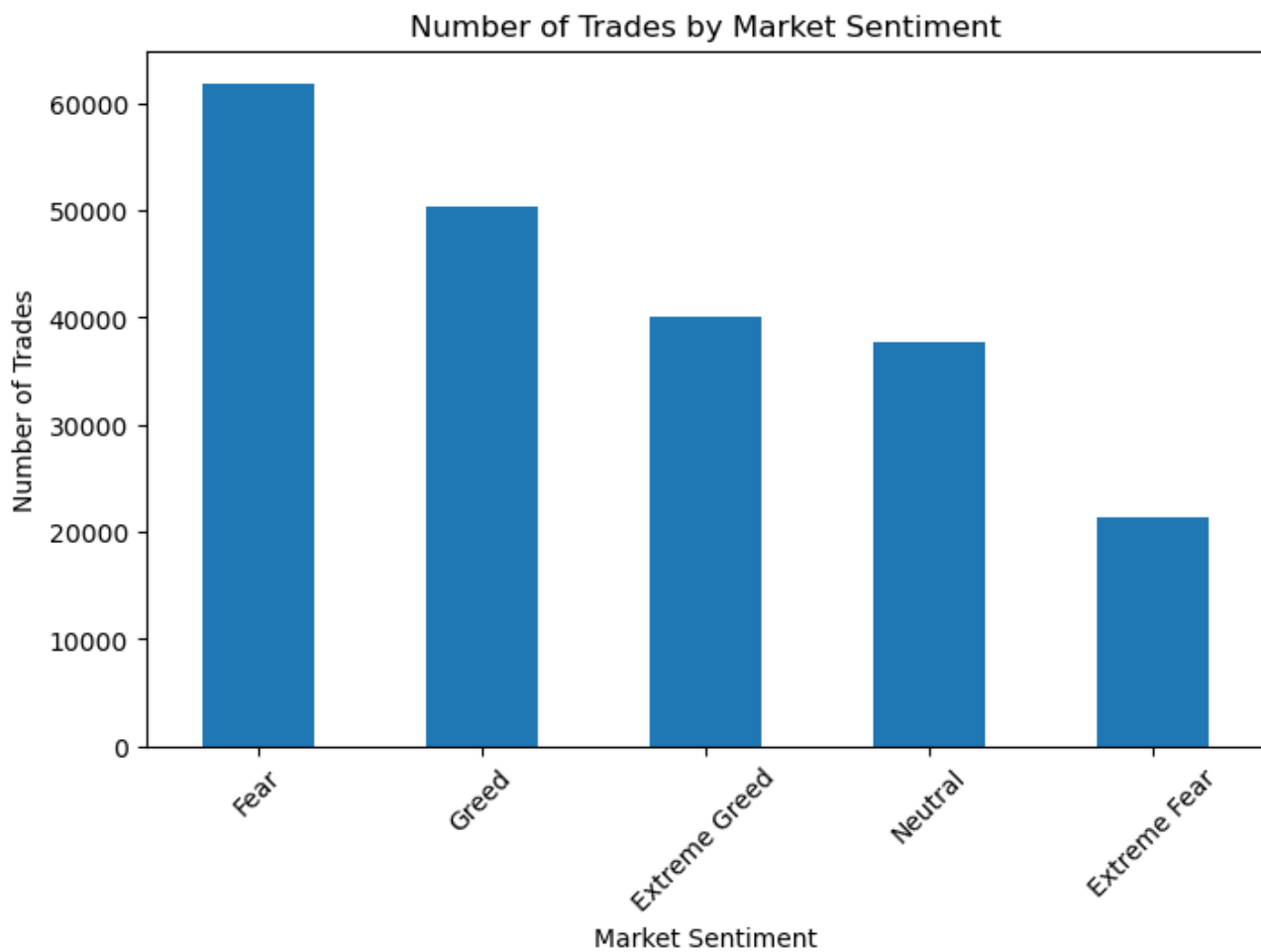
Results & Discussion

1. Trader Profitability Across Market Sentiment



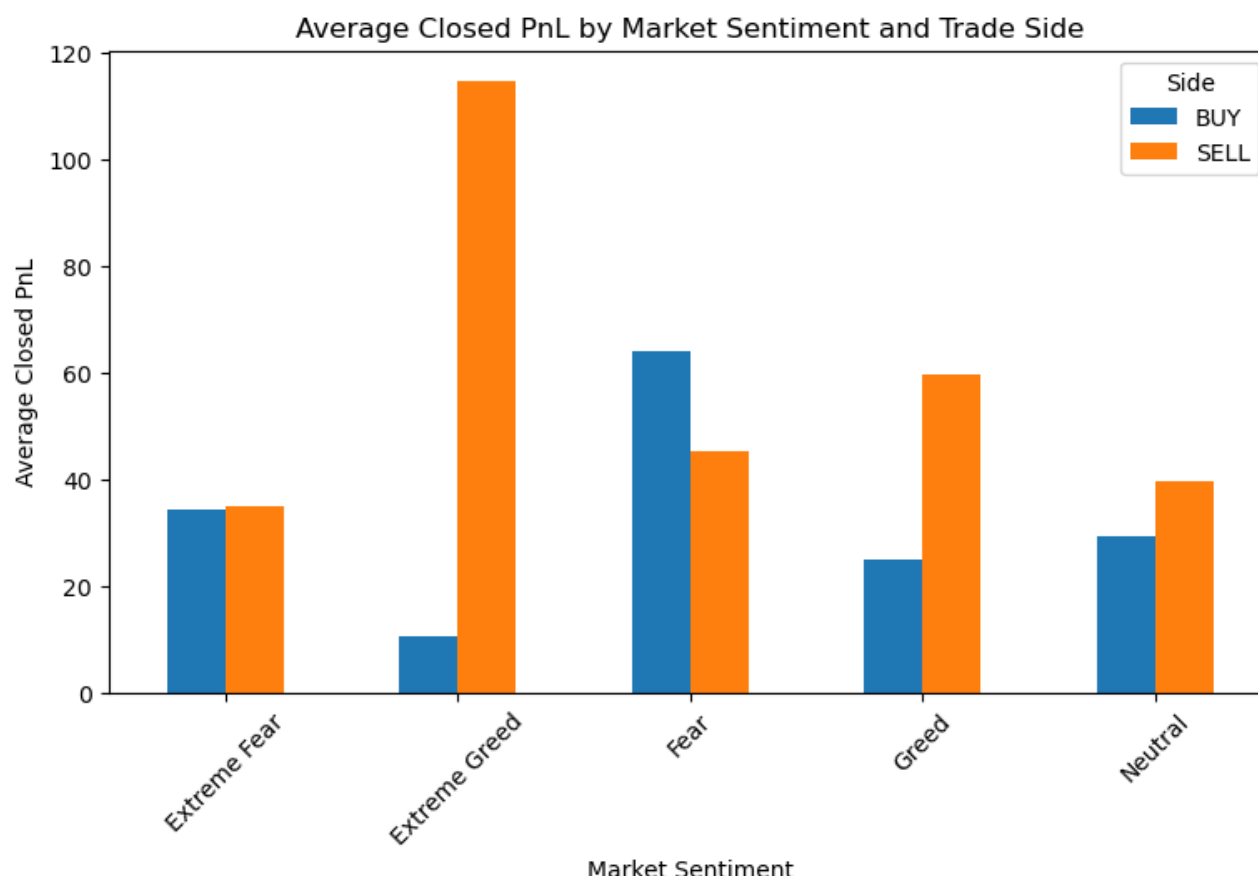
The analysis shows that average trader profitability varied across market sentiment categories. The highest average Closed PnL was observed during Extreme Greed, followed by Fear. In contrast, Neutral and Extreme Fear recorded the lowest average profits. This indicates that trader performance was associated with prevailing market sentiment during the analysis period.

2. Trading Activity Across Market Sentiment



Trading activity was highest during **Fear**, followed by **Greed**. This suggests that traders were more active during bearish market conditions. However, higher trading activity did not necessarily correspond to higher average profits, as **Extreme Greed** recorded the highest average Closed PnL despite fewer trades.

3. BUY vs SELL Performance



Trading direction influenced realized profitability under different market conditions. During **Fear**, BUY trades achieved higher average Closed PnL than SELL trades. Conversely, during **Greed** and particularly **Extreme Greed**, SELL trades generated substantially higher average profits than BUY trades. This highlights that trade direction and market sentiment together were associated with differences in trading outcomes.

Conclusion

This analysis explored the relationship between Bitcoin market sentiment and Hyperliquid trader performance by combining trading records with the Bitcoin Fear & Greed Index.

The results showed that trader behavior and profitability varied across different market sentiment conditions. While **Fear** periods recorded the highest trading activity, **Extreme Greed** produced the highest average realized profits. Additionally, BUY trades performed better during Fear, whereas SELL trades achieved higher average profits during Greed and Extreme Greed.

Overall, the findings indicate that Bitcoin market sentiment is associated with changes in trader activity and realized profitability. However, these observations represent statistical associations rather than causal relationships, and additional market variables should be considered in future analyses.

Recommendations

- Incorporate market sentiment indicators as an additional input when evaluating trading opportunities.
- Evaluate BUY and SELL strategies separately under different market sentiment conditions.

- Apply robust risk management practices during periods of high market volatility due to the presence of significant profit and loss outliers.
- Extend future analyses by incorporating leverage, holding period, market volatility, and trader-level metrics to better understand the factors influencing profitability.